Study Guide

IoT (Internet of Things) using Raspberry Pi and Scratch

Prepared by Ming-Shan Su, Ph.D.

IoT (Internet of Things) using Raspberry Pi and Scratch

Workshop Topics:	Workshops (WS)		
Contents* (IoT)	WS-1	WS-2	WS-3
1. Introducing IoT (Internet of Things), Raspberry Pi, and Scratch 3	✓	✓	✓
2. Game Programming - Scratch 3	✓	✓	✓
3. Exploring Scratch Extensions	✓	✓	✓
4. Getting to Know your Raspberry Pi Take a brief guided tour of your new computer	✓	✓	✓
5. Virtual IT Field Trip with Q & A Choctaw Nation – IT Dept.	✓	✓	✓
6. Special Speaker: The IT Professional A day in the life of an IT Professional and Q&A	✓	✓	✓
7. Starting your Raspberry Pi	✓	✓	✓
8. Hands-on with the Sense HAT Use the sensors and LED matrix display of this add-on board	✓	✓	✓
9. Hands-on IoT Devices with Scratch Use the things: LEDs, Sensors (soil moisture, temperature, humidity, distance, … *), *: depends on the class progress		✓	✓
10. Programming with Python (will be briefly covered).			✓
11. Let's Stargate the Present and the Future	✓	✓	✓

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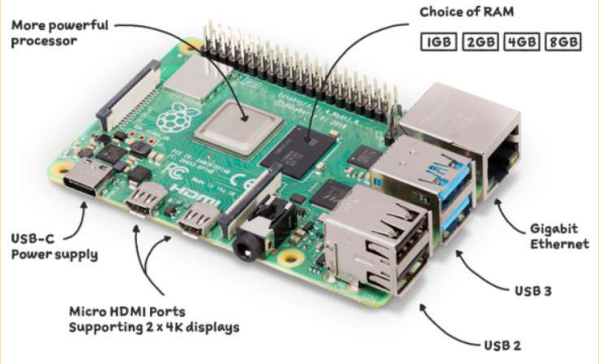
1. Introducing IoT (Internet of Things), Raspberry Pi, and Scratch 3	✓	✓	✓

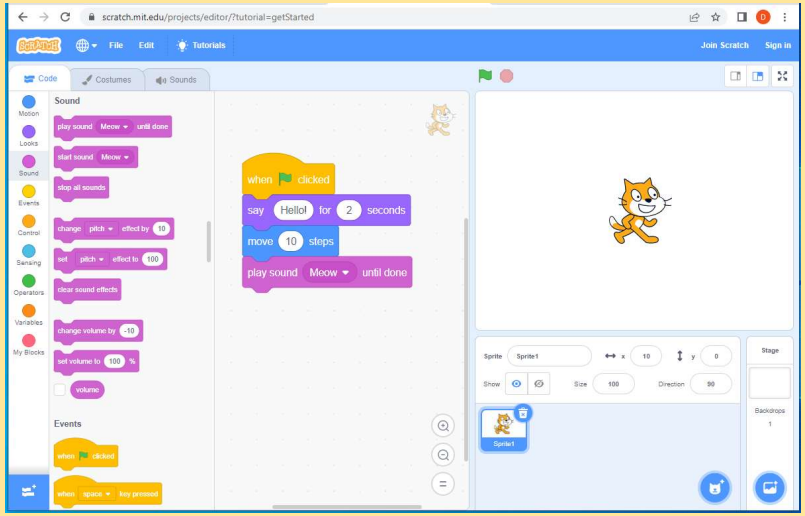
IoT (Internet of Things)

Raspberry Pi



Source: https://www.raspberrypi.com/products/raspberry-pi-4-model-b/

Scratch coding software



Source: https://scratch.mit.edu/

Introduction

- What is Internet of Things (IoT)?
- What is a Raspberry Pi?
- What is Scratch?

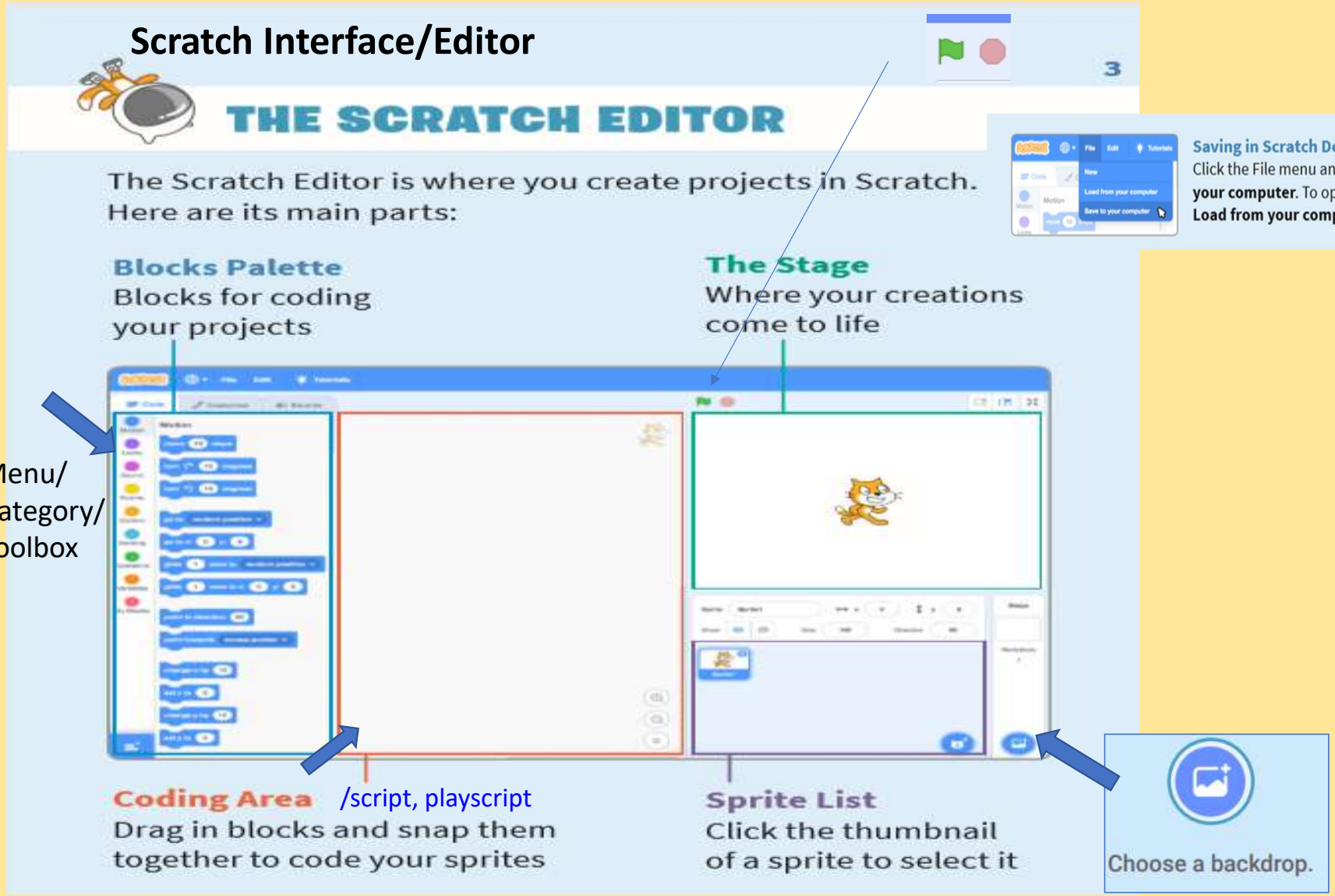
- What is Internet of Things (IoT)?
  - The **Internet of Things** describes physical objects with sensors, processing ability, software, and other technologies that connect and exchange data with other devices and systems over the Internet or other communications networks. Wikipedia
- What is a Raspberry Pi?
  - The **Raspberry Pi** is a tiny and affordable computer that you can use to learn programming through fun, practical projects. – https://ww.raspberrypi.org
- What is Scratch?
  - **Scratch** is a free block-based programming language and online community where you can create your own interactive stories, games, and animations. - https://scratch.mit.edu

https://youtu.be/LlhmzVL5bm8

https://youtu.be/uXUjwk2-qx4

https://youtu.be/jXUZaf5D12A

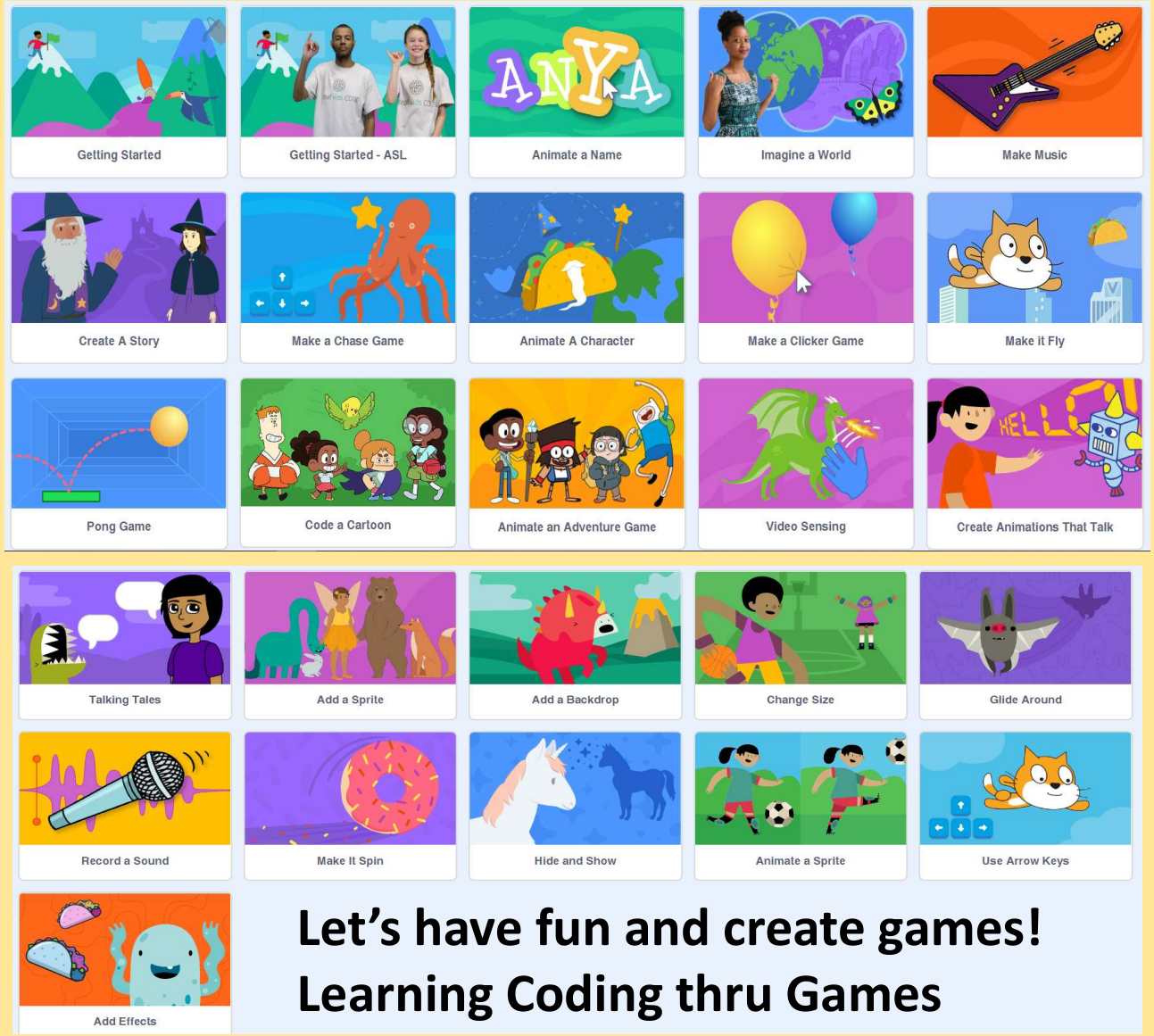
Scratch Interface/Editor



Scratch – the basic concepts of coding

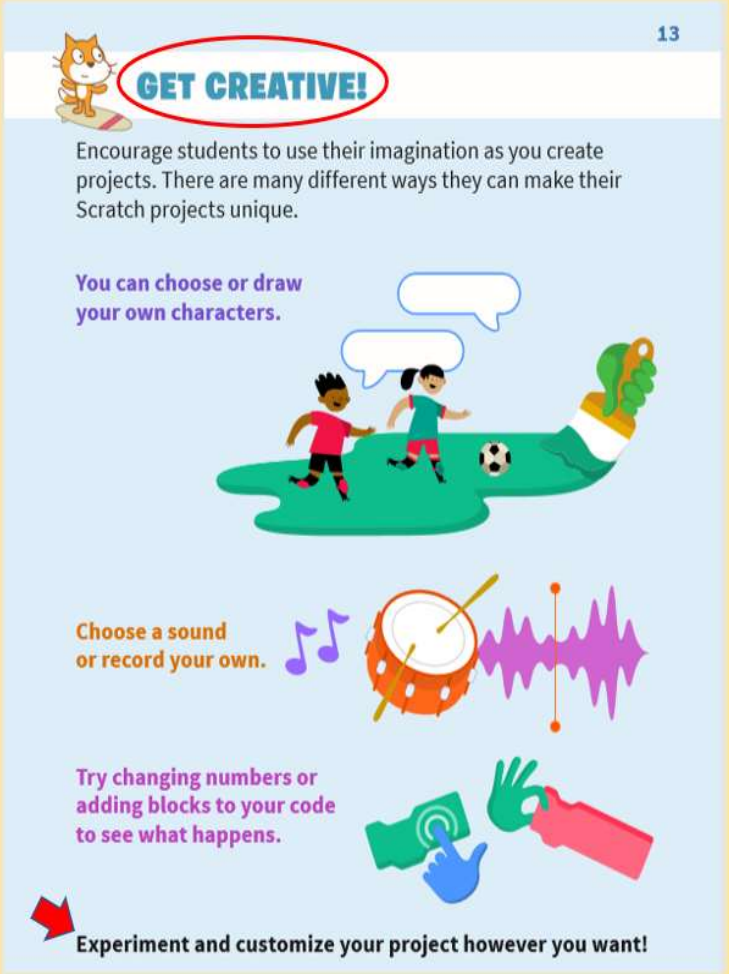
- Basic computer inputs and outputs
- Declaring variables
- Arithmetic Operation (+, -, *, /)
- Control structures
  - Sequence
  - Repetition (Looping/iteration)
  - Selection (Condition)
- Events (key pressed, mouse clicked, sprites overlapping, …)
- Other topics (cloning, broadcast message, ..)

- Creating **hands-on game projects**.



Let's have fun and create games!
Learning Coding thru Games

Pick and Choose the Video Tutorials to familiar with Scratch



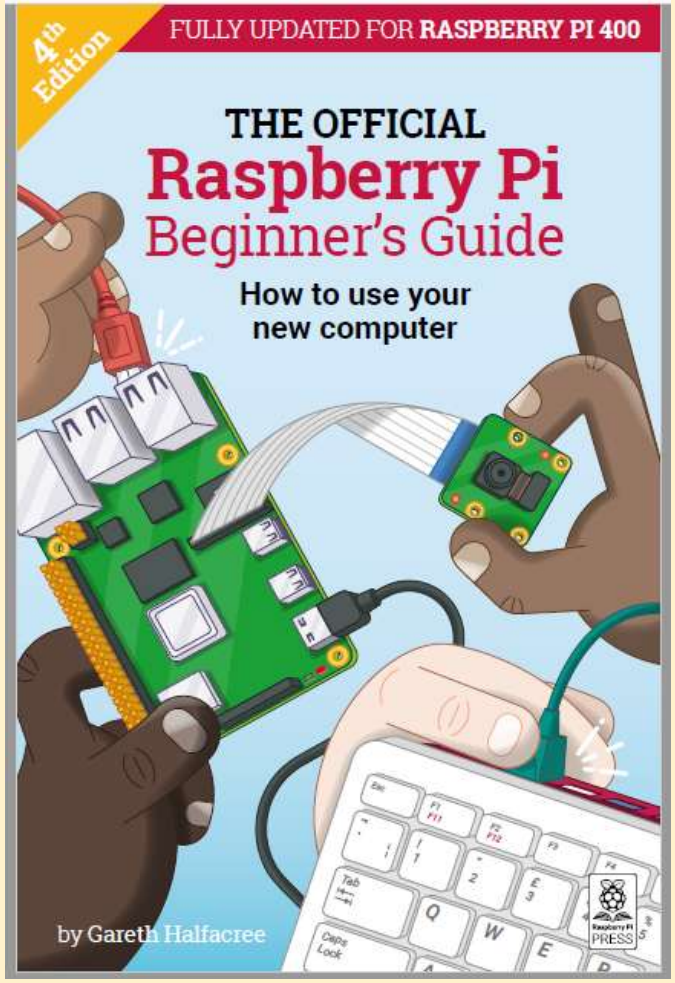
Game options

- Refer to fly-cards.pdf for more information.
- What is next?

Can you design the Pong Game for your family?

Basic to Advanced Scratch Games

Raspberry Pi - Beginner's Guide



https://magpi.raspberrypi.com/books/beginners-guide-4th-ed

Know your IoT sensors & Sense HAT

13. Sensors.....
13.0 Introduction
13.1 Using Resistive Sensors
13.2 Measuring Light 382
13.3 Measuring Temperature with a Thermistor 383
13.4 Detecting Methane 385
13.5 Measuring CO_2 Concentration 388
13.6 Measuring a Voltage 391
13.7 Reducing Voltages for Measurement 394
13.8 Using Resistive Sensors with an ADC 396
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13.10 Measuring the Raspberry Pi CPU Temperature 400
13.11 Measuring Temperature, Humidity, and Pressure with a Sense HAT 402
13.12 Measuring Temperature Using a Digital Sensor 404
13.13 Measuring Acceleration with an MMA8452Q Module 407
13.14 Finding Magnetic North with the Sense HAT 411
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13.17 Sensing a Magnet with the Sense HAT 416
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13.20 Capacitive Touch Sensing 422
13.21 Reading Smart Cards with an RFID Reader/Writer 425
13.22 Displaying Sensor Values 428
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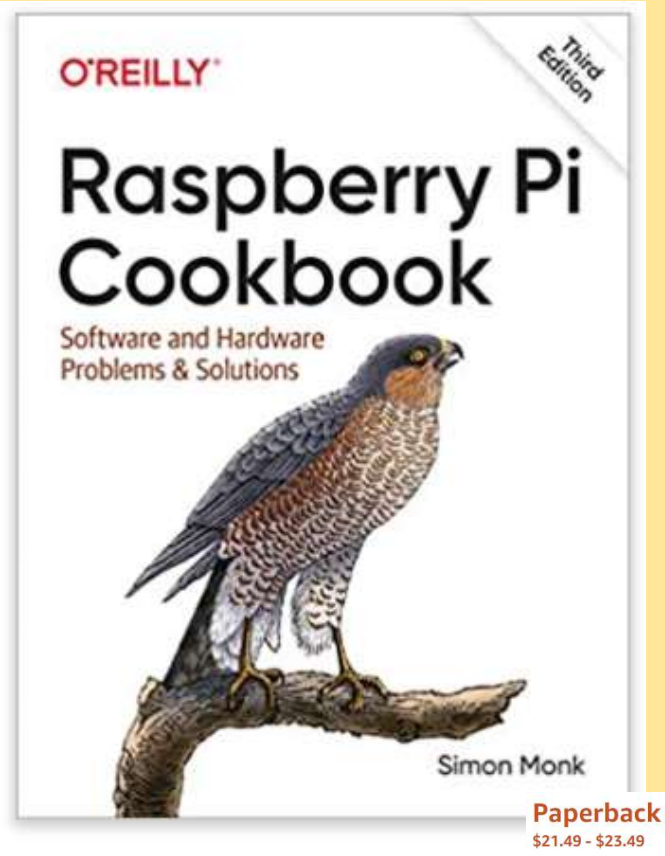

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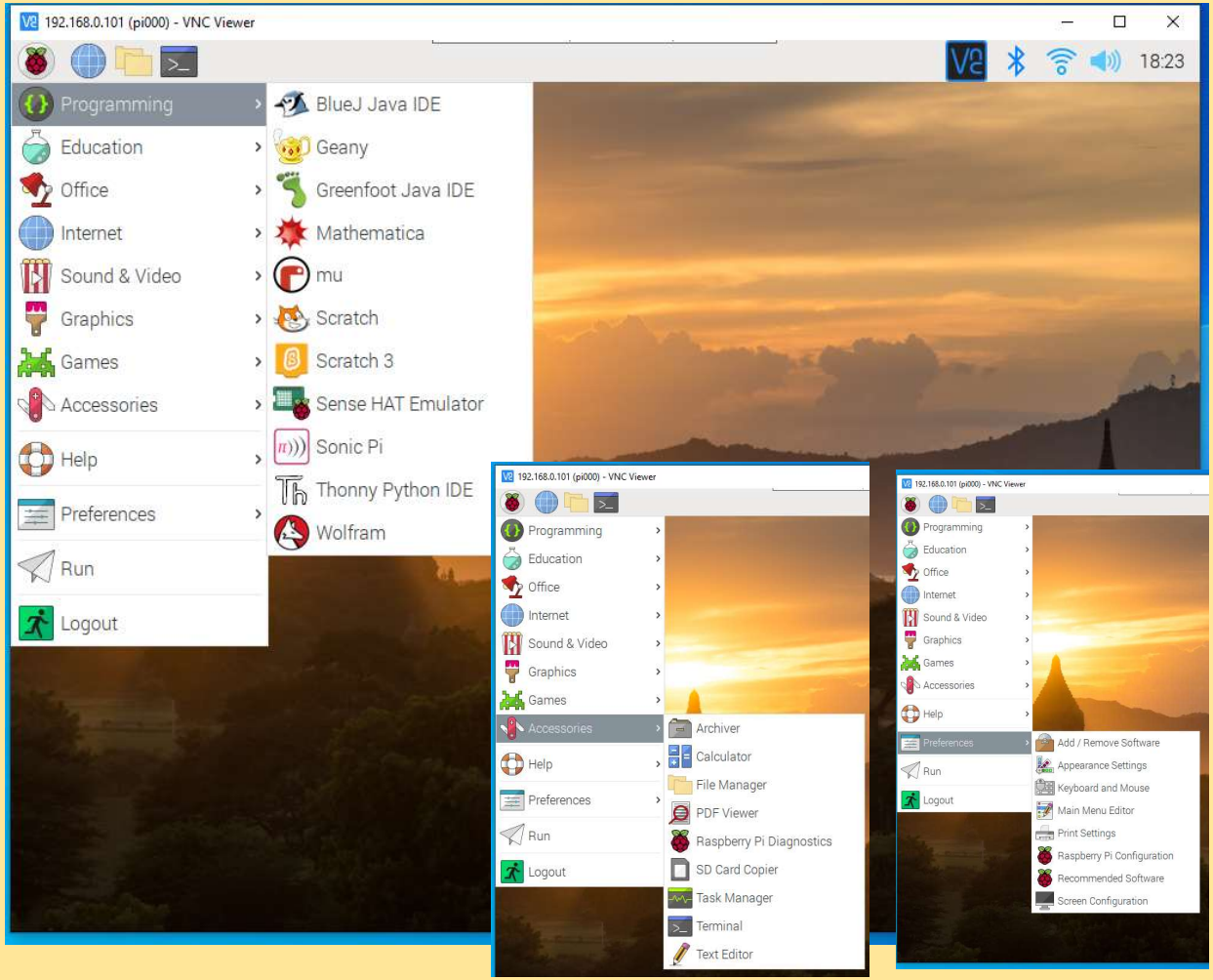
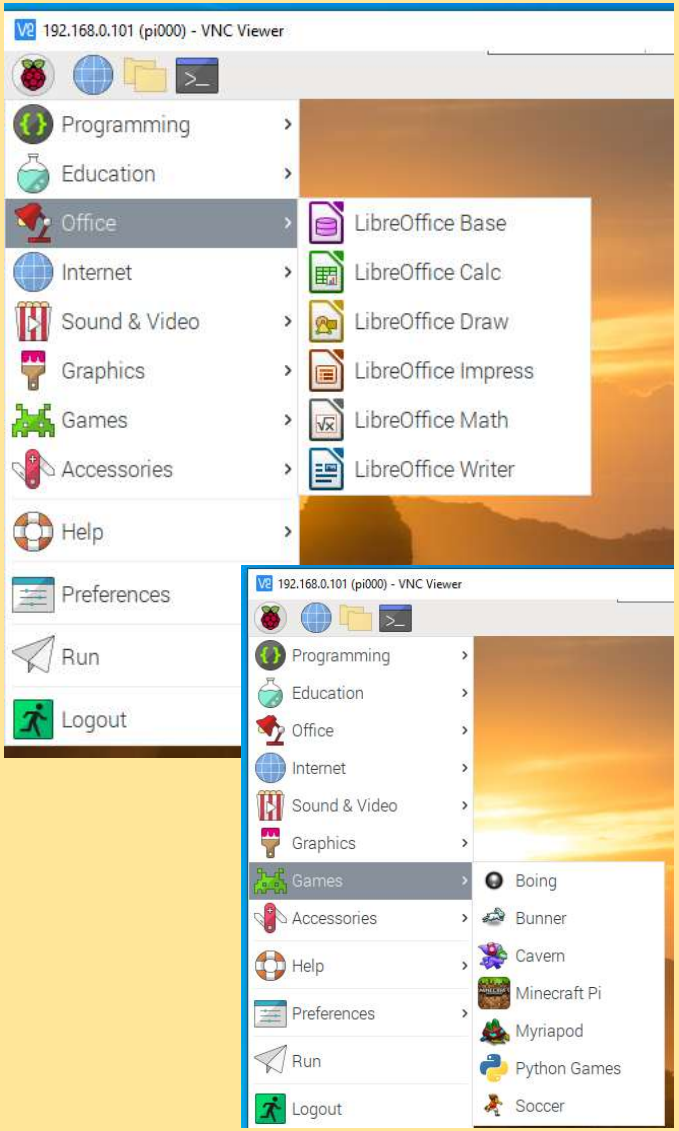
https://www.amazon.com/Raspberry-Pi-Cookbook-Software-Solutions/dp/1492043222/ref=asc_df_1492043222/?tag=hyprod-20&linkCode=df0&hvadid=385577217796&hvpos=&hvnetw=g&hvrand=12960532541917037430&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=9026684&hvtargid=pla-546274382606&psc=1&tag=&ref=&adgrpid=79288121195&hvpone=&hvptwo=&hvadid=385577217796&hvpos=&hvnetw=g&hvrand=12960532541917037430&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=9026684&hvtargid=pla-546274382606

What you have on Raspberry Pi - all the free software/apps!





The projects combining Raspberry Pi
+ Sense HAT

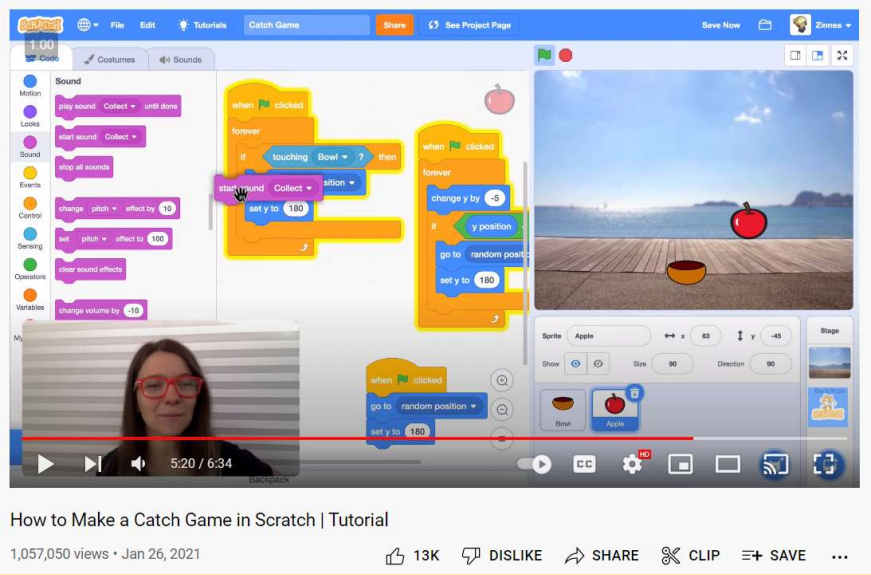
3 Simple Raspberry Pi Sense HAT Projects for Kids and Families

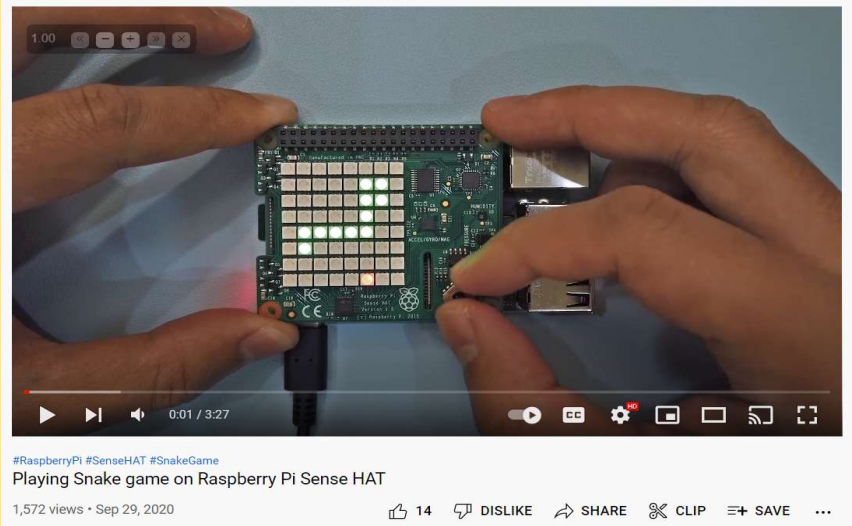
By Les Pounder published May 03, 2020

https://www.tomshardware.com/how-to/raspberry-pi-sense-hat-projects-for-kids

Extra game options

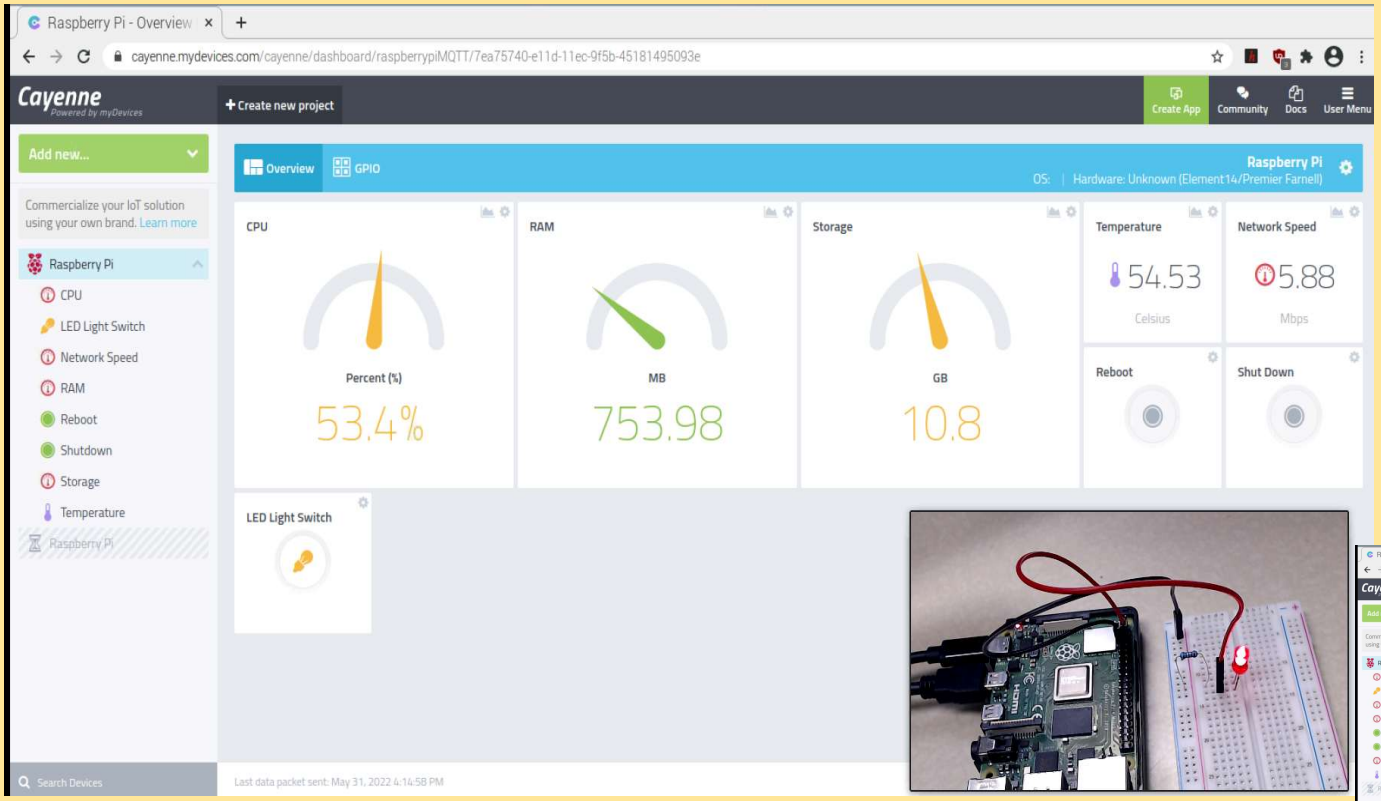
- How to Make a Catch Game in Scratch | Tutorial
  - https://youtu.be/7NN5v2wSL4U [6:34]
- Playing Snake game on Raspberry Pi Sense HAT
  - https://youtu.be/jTNnlyIc46g [3:27]
  - https://youtu.be/jTNnlyIc46g?t=74
  - Code, https://codereview.stackexchange.com/questions/234969/snake-game-for-raspberry-pi-sense-hat-written-in-python



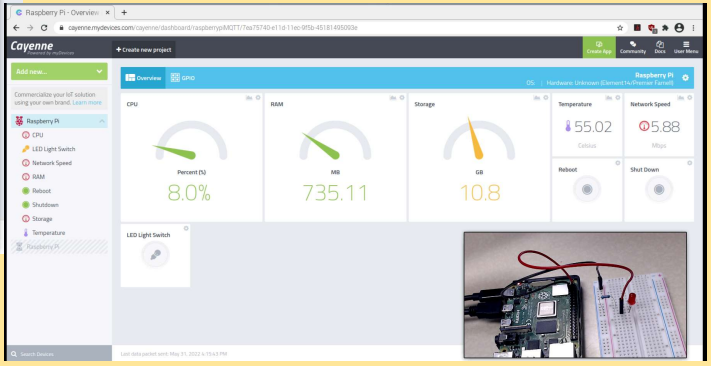


IoT application: Put them all together! (Scratch, Raspberry Pi, IoT-Things, …)

Cayenne IoT (Internet of Things) Platform



That is all for Workshop-1.



https://youtu.be/ovthX9QlVF8?t=7 [4m:03s]